4A
CHEMICAL REACTIONS

UNIT 01: FOUNDATIONAL CHEMISTRY SKILLS

CHEMISTRY 1310

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LEARNING OBJECTIVES

CHAPTER 4

1. Write complete and balanced chemical equations, based on either chemical symbols or word descriptions, including appropriate phases.
2. Recognize and describe the five basic types of chemical reactions.
3. List and describe the driving forces for reactions in aqueous solution.
4. Describe and write equations representing the dissociation of ionic compounds upon dissolution in water.
5. Define and apply the terms strong electrolyte, weak electrolyte, and nonelectrolyte.
6. Identify compounds as being strong electrolytes, weak electrolytes, or nonelectrolytes from chemical formulas.
7. Apply solubility guidelines to predict the formation of a precipitate in reactions of ionic compounds in aqueous solution.
8. Write and interpret ionic and net ionic equations for precipitation reactions.

Chemical Equations

4.1

Chemical equations

- Designed to represent the transformation of one or more chemical species into new substances.

Reactants: The starting components of a reaction

Products: The ending components.

Chemical equations are always written:

$$\text{Reactants} \rightarrow \text{Products}$$

Chemical Equations

4.1

A great deal of information can be gleaned from a chemical equation:

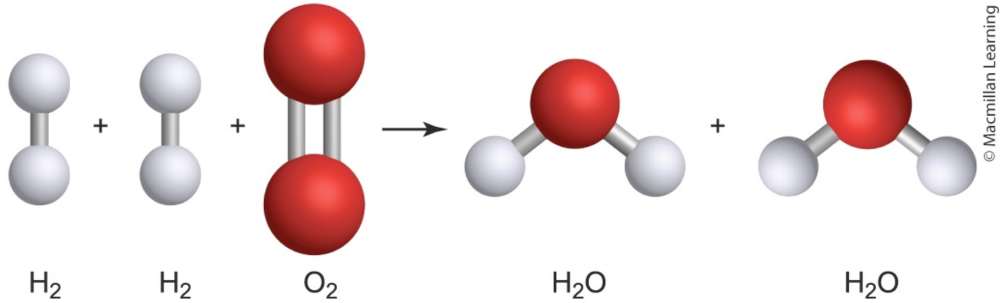

FIGURE 4.1

TABLE 4.1

Information	Notation	Example
What happens?	Identity (names or formulas) of the reactants and products	Hydrogen (H_2) and oxygen (O_2) react to form water (H_2O).
In what proportions?	Coefficients placed before the formulas	$2\ H_2 + O_2 \rightarrow 2\ H_2O$
In what physical states?	(*s*), (*l*), (*g*), (*aq*) included after the formulas	$2\ H_2(g) + O_2(g) \rightarrow 2\ H_2O(l)$
Special conditions?	Special conditions written above or below the arrow	$\xrightarrow{\text{Heat}}$

Chemical Equations

4.1

In a balanced chemical equation, the number of atoms of *each element* on the left (reactants) side of the chemical equation is **equal** to the number of atoms of that element on the right (products) side of the chemical equation.

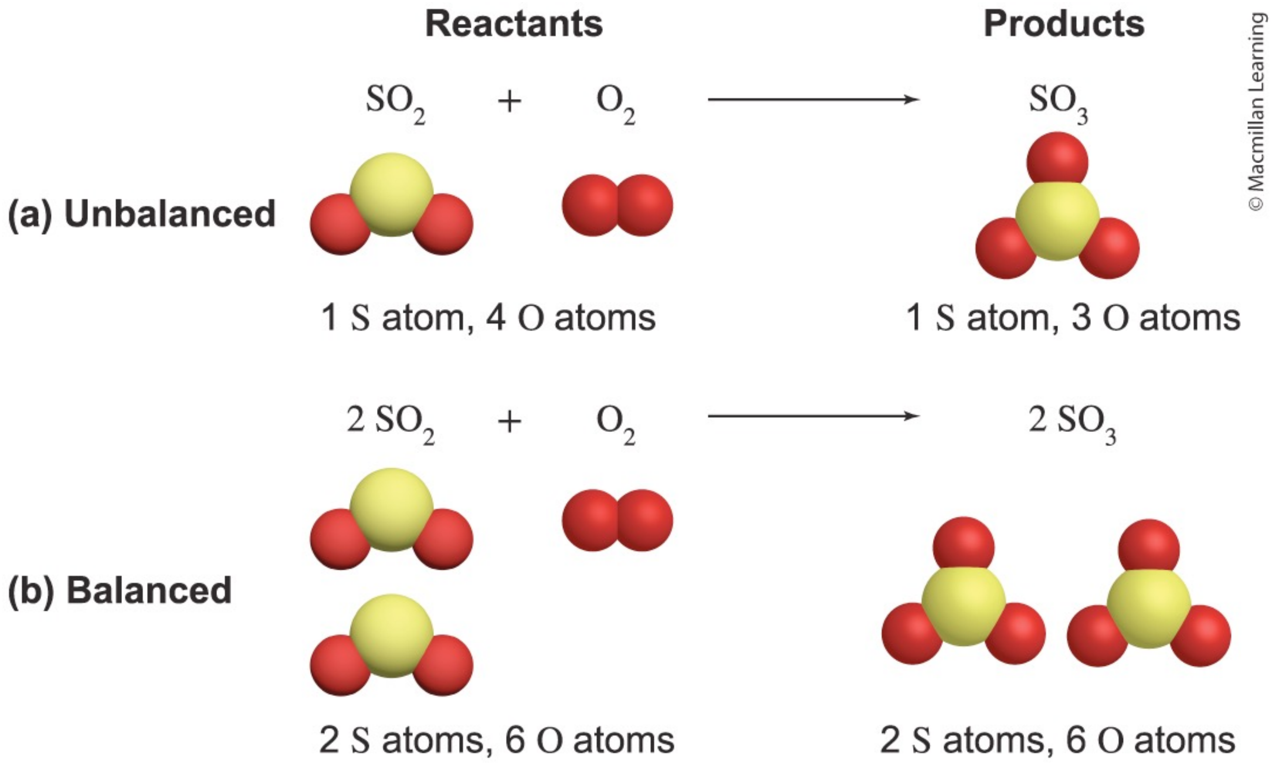

FIGURE 4.3

Chemical Equations

4.1

Chemical equations can be balanced using this four-step strategy:

1. Balance polyatomic ions as whole units if they are present as both reactants and products.
2. Then, balance elements that appear in only one reactant and one product.
3. Then, balance any remaining elements; resolve odd/even issues by doubling previously determined coefficients.
4. Verify that the number of each type of atom is the same in the reactants and products and that the coefficients are in the simplest form (smallest whole number coefficients).

EXAMPLE PROBLEM 1

Learning Objective(s): 4.1

Balance the reaction using the smallest set of whole number coefficients.

TIP Don't start with O

$$\underline{1}\ C_3H_8\,(g) + \underline{5}\ O_2\,(g) \longrightarrow \underline{3}\ CO_2\,(g) + \underline{4}\ H_2O\,(g)$$

TIP standard: reduce to smallest set of coefficients

TIP check your work at end by counting atoms

reactants	products
3 C atoms	3 C atoms
8 H atoms	8 H atoms
10 O atoms	10 O atoms

EXAMPLE PROBLEM 2

Learning Objective(s): 4.1

diatomic O_2!

Ammonia (NH_3) reacts with oxygen gas to form water and nitrogen dioxide.

Write a balanced chemical equation using the smallest whole number coefficients.

Start with H!

$$4NH_3 + \cancel{7}O_2 \longrightarrow 6H_2O + 4NO_2$$

N	$\cancel{1}$	4
H	$\cancel{3}$	6
O	$\cancel{2}$	14

N	$\cancel{1}$	4
H	$\cancel{2}$	6
O	$\cancel{3}$	14

IN-CLASS QUESTION 1

Learning Objective(s): 4.1

What is the coefficient for each chemical species when the following reaction is balanced with the smallest possible whole number coefficients?

$$\underline{\ 1\ }\ Al_2(CO_3)_3 + \underline{\ 2\ }\ H_3PO_4 \longrightarrow \underline{\ 2\ }\ AlPO_4 + \underline{\ 3\ }\ CO_2 + \underline{\ 3\ }\ H_2O$$

start with Al, C, or H

easier to balance polyatomic as <u>single unit</u>

Types of Chemical Reactions

4.2

In this chapter, we'll explore five primary types of chemical reactions:

1. Synthesis reactions
2. Decomposition reactions
3. Single-replacement reactions
4. Double-replacement reactions
5. Combustion reactions

opposities

Types of Chemical Reactions

4.2

Synthesis reactions (or combination reactions) start with simple reactants that combine to form a single more complex product.

$$4\ Fe(s) + 3\ O_2(g) \rightarrow 2\ Fe_2O_3(s)$$

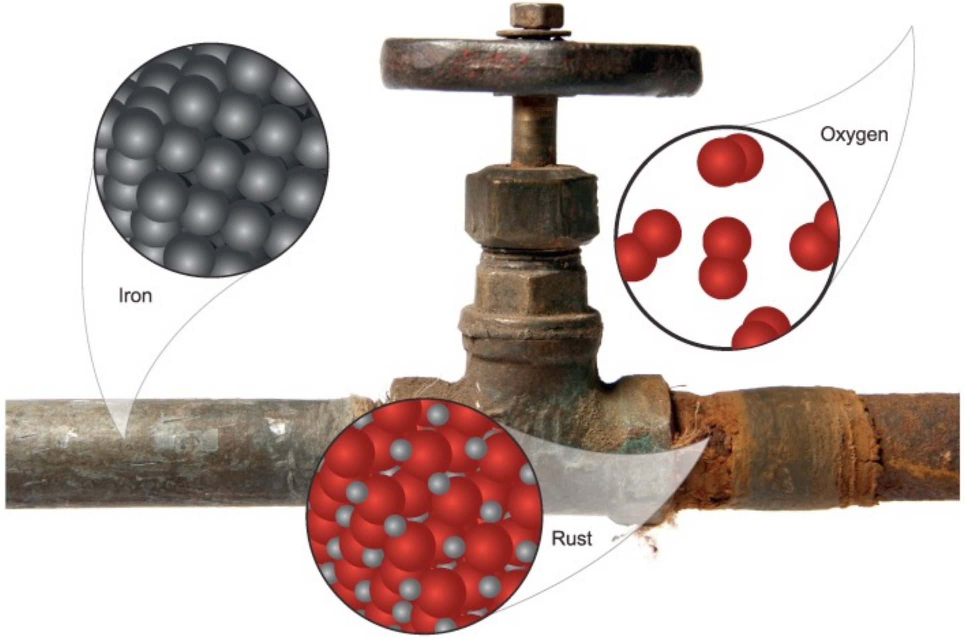

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FIGURE 4.6 Synthesis Reaction of Rust on Iron

Types of Chemical Reactions

4.2

Decomposition reactions occur when a single reactant breaks down into less complex products.

$$2\ H_2O(l) \rightarrow 2\ H_2\ (g) + O_2(g)$$

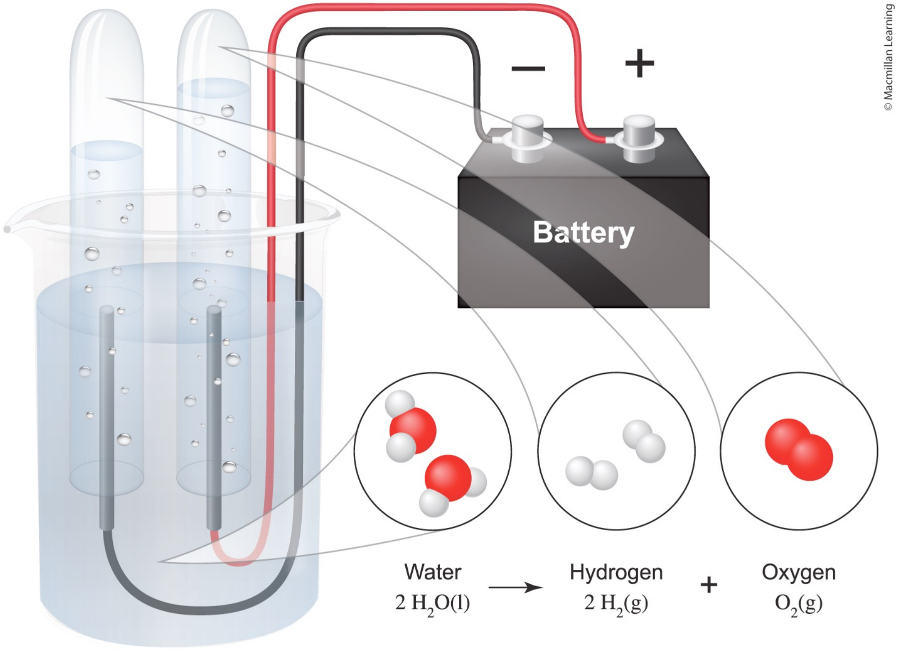


FIGURE 4.7 Decomposition Reaction: Electrolysis of Water

Types of Chemical Reactions

4.2

Single-replacement reactions (also called single-displacement reactions) occur when an element reacts with a compound and displaces one of the elements in that compound, producing a new compound and a new element.

$$Zn(s) + 2\,HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$$

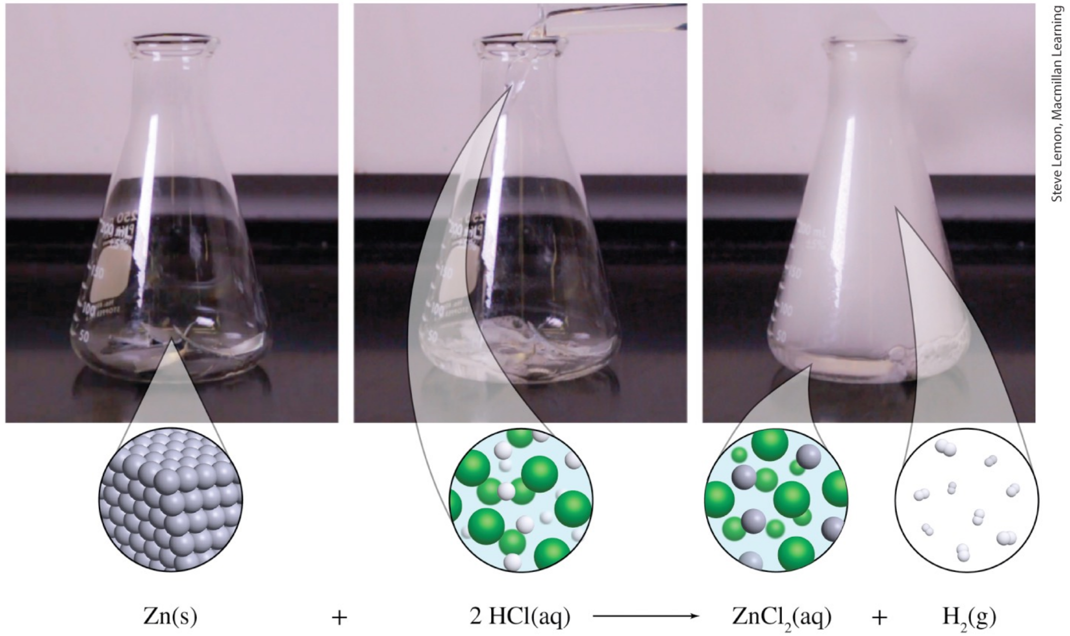


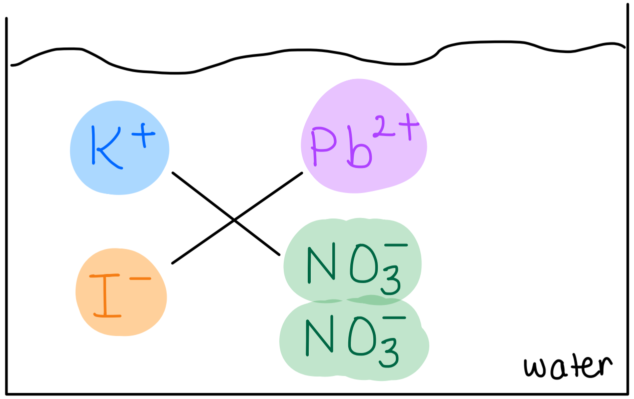
FIGURE 4.10 Single-Replacement Reaction of Zn with HCl

Types of Chemical Reactions

4.2

Double-replacement reactions (also called double-displacement or metathesis reactions) occur when two ionic compounds exchange their constituent ions (cations and anions) to form two new ionic compounds.

$$2\ KI(aq) + Pb(NO_3)_2(aq) \longrightarrow PbI_2(s) + 2\ KNO_3(aq)$$



* pair up cation + anion to form new neutral ionic compounds *

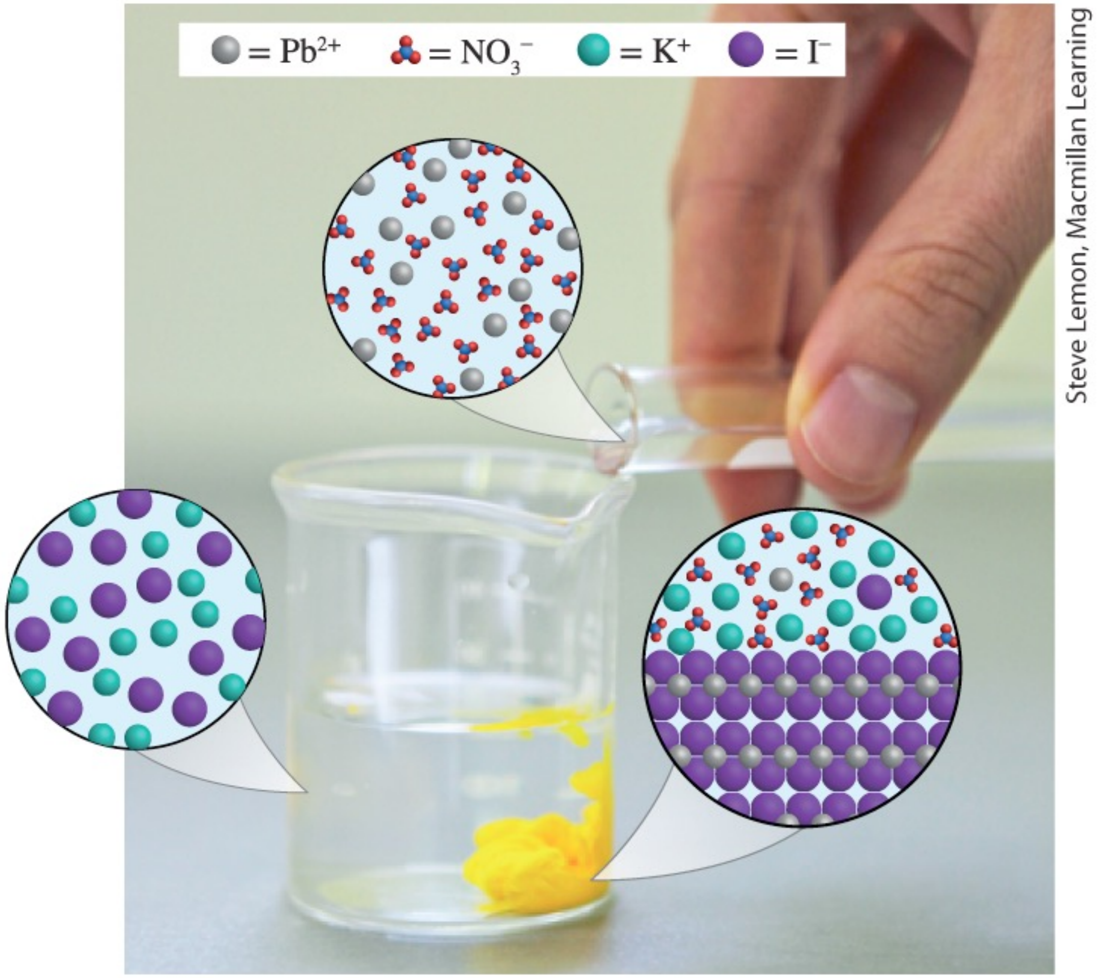


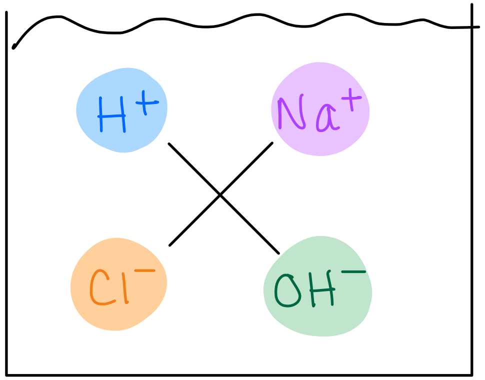
FIGURE 4.12 Double-Replacement Reaction of KI with $Pb(NO_3)_2$

Types of Chemical Reactions

4.2

Acid-base reactions are a special case of double-replacement reactions where water (H_2O) is always a product.

- An acid is a type of compound usually written with H at the beginning of its formula.
- A base is a type of compound usually written with OH at the end of its formula (e.g. hydroxide compounds).



$$\underbrace{HCl(aq)}_{\text{acid}} + \underbrace{NaOH(aq)}_{\text{base}} \rightarrow NaCl(aq) + \underbrace{H_2O(l)}_{\text{water}}$$

$H^+ OH^- = H_2O$

$$H_2SO_4(aq) + 2\ KOH(aq) \rightarrow K_2SO_4(aq) + 2\ H_2O(l)$$

Types of Chemical Reactions

4.2

Combustion reactions occur when a substance undergoes rapid combination with oxygen (O_2).

- If the substance is a hydrocarbon (a molecular/covalent compound comprised of C, H, and sometimes O), then the products will always be carbon dioxide (CO_2) and water (H_2O).

$$C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O(l)$$

$$2\ C_{24}H_{50}(s) + 73\ O_2(g) \rightarrow 48\ CO_2(g) + 50\ H_2O(l)$$

Types of Chemical Reactions

4.2

TABLE 4.2 Summary of Reaction Types

Reaction Type	Generic Formula	Examples
Synthesis	$A + B \rightarrow AB$	$4\ Fe(s) + 3\ O_2(g) \rightarrow 2\ Fe_2O_3(s)$
Decomposition	$AB \rightarrow A + B$	$2\ H_2O(l) \rightarrow 2\ H_2(g) + O_2(g)$ $2\ KClO_3(s) \rightarrow 2\ KCl(s) + 3\ O_2(g)$
Single-Replacement	$A + BC \rightarrow AC + B$	$Zn(s) + 2\ HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$
Double-Replacement	$AB + CD \rightarrow AD + CB$	$2\ KI(aq) + Pb(NO_3)_2(aq) \rightarrow PbI_2(s) + 2\ KNO_3(aq)$ $HBr(aq) + KOH(aq) \rightarrow KBr(aq) + H_2O(l)$
Combustion	$C_xH_y + O_2 \rightarrow CO_2 + H_2O$	$C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O(l)$

IN-CLASS QUESTION 2

Learning Objective(s): 4.2

Label the type of chemical reaction using the terms: synthesis, decomposition, single-replacement, or double-replacement.

complex

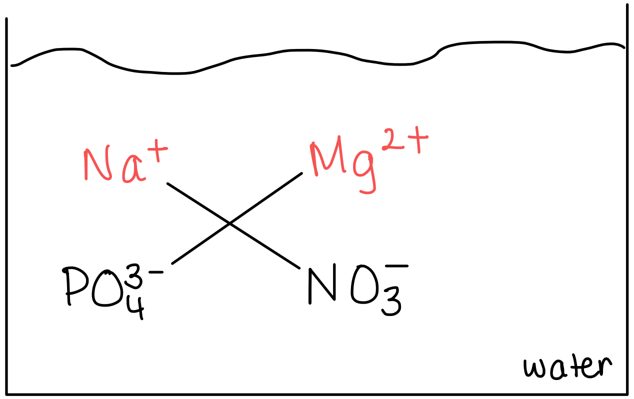
$2\ HgO(s) \rightarrow 2\ Hg(l) + O_2(g)$ decomposition

$2\ NO(g) + O_2(g) \rightarrow 2\ NO_2(g)$ synthesis

$2\ Na(s) + 2\ H_2O(l) \rightarrow 2\ NaOH(aq) + H_2(g)$ single replacement, Na displaces H

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ decomposition

$2\ Na_3PO_4(aq) + 3\ Mg(NO_3)_2(aq) \rightarrow Mg_3(PO_4)_2(aq) + 6\ NaNO_3(aq)$ double replacement



Types of Chemical Reactions

4.2

Reactions occur spontaneously due to a combination of changes in heat energy and randomness, which are discussed in future chapters.

Certain patterns of reactants and products tend to be associated with spontaneous reactions and can be classified by their **driving force**:

1. Formation of a precipitate double replacement rxns
2. Formation of water or neutralization (discussed in Chapter 16) acid-base rxns
3. Oxidation and reduction (discussed in Chapter 19) combustion, synthesis, decomposition, single replacement rxns

IN-CLASS QUESTION 3

Learning Objective(s): 4.3

Identify the driving force for each reaction:

BASE ACID WATER

$Mg(OH)_2(aq) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + 2\ H_2O(l)$

formation of liquid water
acid-base (neutralization)

INSOLUBLE

$K_2CO_3(aq) + CaBr_2(aq) \rightarrow CaCO_3(s) + 2\ KBr(aq)$

formation of a solid
double replacement (precipitation)

$2\ HCl(aq) + Mg(s) \rightarrow MgCl_2(aq) + H_2(g)$

reduction-oxidation
single replacement rxns